

CE 528 : Design of Masonry Structures

Course Code : CE 528

Course Name : Design of Masonry Structures

L-P-T-C: 3-0-0-3

Intended for: UG and PG

Prerequisites: CE 351 – Design of Reinforced Concrete Structures

Mutual Exclusion: NA

Approval : 57th BoA

Course Contents

- **Historical Perspective of Masonry Structures:** Different types of masonry structures (Domes, Buildings, Bridges, Arches, etc.); Traditional/Heritage masonry structures; Foundations and site selection; Behavior of masonry structures during past earthquakes: Common modes of failure, effect of unit shapes and mortar type, effect of roof and floor systems, Common deficiencies. [6 hours]
- **Masonry Properties:** Masonry units- stones, brick and concrete blocks, hollow and solid units; Manufacturing process; Mortar, grout and reinforcement; Various tests and standards, Stress-Strain Characteristics [6 hours]
- **Masonry Under Compression:** Prism strength, Failure mechanism, types of construction and bonds; Eccentric loading; Slenderness – effective length and effective height, effect of openings; Code provisions. [8 hours]
- **Masonry Under Lateral Loads:** In-plane and out-of-plane loads, bending parallel and perpendicular to bed joints; Shear and flexure behavior of piers; Test and standards; Analysis of perforated shear walls, lateral force distribution for flexible and rigid diaphragms; Arching action; Combined axial and bending actions; Masonry infills: Effect of masonry infills on seismic behavior of framed buildings, Failure modes, Code provisions. [10 hours]
- **Earthquake Resistant Measures:** Analysis for earthquake forces, role of floor and roof diaphragm; Concept and design of bands, bandages, splints, and ties; Reinforced masonry; Vertical reinforcement at corners and jambs; Measures in random-rubble masonry; Introduction to confined masonry [8 hours]
- **Retrofitting of Masonry Building:** Techniques of repair and retrofitting of masonry buildings; IS: 13935-1993 provision for retrofitting. [4 hours]

Text books:

1. MJN Priestley and T Paulay, **Seismic design and assessment of reinforced concrete and masonry buildings**, John Wiley and Sons, 1997.
2. RG Drysdale, AA Hamid, LR Baker, **Masonry Structures: Behaviour and Design**, Prentice Hall, 1994.

References:

1. AW Hendry, **Structural Brickwork**, The Macmillan Press Ltd., 1981.

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2. R E Klingner, **Masonry structural design**, McGrawHill Companies, Inc., 2010.
 3. M Tomazevic, **Earthquake-resistant design of masonry buildings**, Series on Innovation in Structures and Construction, Vol. 1, Imperial College Press, 1999.
 4. National Building Code of India 2016 Vol.1, Part 6 Section 4 Structural Design - Masonry
 5. IS 1905: 1987, Handbook on Masonry Design and Construction.
 6. IS 17848 : 2022: Confined Masonry For Earthquake Resistance Code of Practice